

Module 3: Functions in JavaScript

A **Function** is a block of code designed to perform a specific task. Functions help make code reusable, organized, and easier to maintain.

1. Function Declaration

A Function Declaration defines a function using the function keyword.

Syntax

```
function functionName() {  
    // code  
}
```

Example

```
function greet() {  
    console.log("Welcome to JavaScript");  
}
```

```
greet();
```

Output

Welcome to JavaScript

Practical Example: Calculate Area of Rectangle

```
function rectangleArea(length, width) {  
  let area = length * width;  
  console.log("Area =", area);  
}
```

```
rectangleArea(10, 5);
```

Output

Area = 50

2. Parameters

Parameters are variables listed inside the function definition.

Syntax

```
function functionName(parameter1, parameter2) {  
  // code  
}
```

Example

```
function add(a, b) {  
  console.log(a + b);  
}
```

```
add(10, 20);
```

Output

30

Practical Example: Student Information

```
function student(name, course) {  
  console.log("Name:", name);  
  console.log("Course:", course);  
}
```

```
student("Vicky", "BCA");
```

Output

Name: Vicky
Course: BCA

3. Return Statement

The return statement sends a value back from a function.

Example

```
function multiply(a, b) {  
  return a * b;  
}
```

```
let result = multiply(5, 4);
```

```
console.log(result);
```

Output

20

Practical Example: Percentage Calculator

```
function percentage(totalMarks, obtainedMarks) {  
    return (obtainedMarks / totalMarks) * 100;  
}
```

```
let per = percentage(500, 420);
```

```
console.log("Percentage =", per + "%");
```

Output

Percentage = 84%

4. Arrow Functions

Arrow Functions provide a shorter syntax for writing functions.

Syntax

```
const functionName = () => {  
    // code  
};
```

Example 1

```
const greet = () => {  
    console.log("Hello Students");  
};
```

```
greet();
```

Output

Hello Students

Example 2: Addition

```
const add = (a, b) => {  
  return a + b;  
};  
  
console.log(add(10, 20));
```

Output

30

Short Form Arrow Function

```
const square = num => num * num;  
  
console.log(square(5));
```

Output

25

Practical Example: GST Calculator

```
const calculateGST = amount => amount * 0.18;  
  
console.log("GST =", calculateGST(1000));
```

Output

GST = 180

Function vs Arrow Function

Function	Arrow Function
Uses function keyword	Uses =>
Longer syntax	Short syntax
Can be called before declaration	Cannot be called before declaration
Has its own this	Inherits this

Comparison Example

```
function sum1(a, b) {  
  return a + b;  
}
```

```
const sum2 = (a, b) => a + b;
```

```
console.log(sum1(5, 5));  
console.log(sum2(5, 5));
```

5. Callback Functions

A Callback Function is a function passed as an argument to another function.

Example

```
function greet(name) {  
  console.log("Hello " + name);  
}
```

```
function processUser(callback) {  
  callback("Vicky");  
}
```

```
processUser(greet);
```

Output

Hello Vicky

Practical Example: Online Order

```
function orderReady() {  
  console.log("Your Pizza is Ready!");  
}
```

```
function placeOrder(callback) {  
  console.log("Preparing Pizza...");  
  callback();  
}
```

```
placeOrder(orderReady);
```

Output

Preparing Pizza...
Your Pizza is Ready!

Practical Example: Calculator with Callback

```
function calculate(a, b, operation) {  
  return operation(a, b);  
}
```

```
function add(x, y) {  
  return x + y;  
}
```

```
function multiply(x, y) {  
  return x * y;  
}
```

```
console.log(calculate(10, 20, add));  
console.log(calculate(10, 20, multiply));
```

Output

30
200

Callback Using Arrow Function

```
function calculate(a, b, callback) {  
  return callback(a, b);  
}
```

```
let result = calculate(10, 5, (x, y) => x - y);
```

```
console.log(result);
```

Output

Real-Life Analogy of Callback Function

Imagine you order food online:

1. You place an order.
2. Restaurant prepares the food.
3. Delivery starts after preparation.

```
placeOrder(prepareFood);  
prepareFood(deliverFood);
```

Here each next action waits for the previous one to finish.
This is the idea behind callbacks.

Comprehensive Practice Program

```
function calculate(a, b, callback) {  
  return callback(a, b);  
}
```

```
const add = (x, y) => x + y;  
const subtract = (x, y) => x - y;  
const multiply = (x, y) => x * y;  
const divide = (x, y) => x / y;
```

```
console.log("Addition:", calculate(20, 10, add));  
console.log("Subtraction:", calculate(20, 10, subtract));  
console.log("Multiplication:", calculate(20, 10, multiply));  
console.log("Division:", calculate(20, 10, divide));
```

Output

Addition: 30

Subtraction: 10

Multiplication: 200

Division: 2

Summary

Topic	Purpose
Function Declaration	Creates a function
Parameters	Receive input values
Return	Sends value back
Arrow Function	Shorter function syntax
Callback Function	Function passed as an argument
Practical Use	Calculators, Events, APIs, Timers, Data Processing

Interesting & Practical Questions on JavaScript Functions

1. Function Declaration

Q1. Create a function to check whether a number is Even or Odd.

```
function checkEvenOdd(num) {  
  if (num % 2 === 0)  
    return "Even";  
  else  
    return "Odd";  
}
```

```
}
```

```
console.log(checkEvenOdd(7));
```

Output

Odd

Q2. Create a function to find the largest of three numbers.

```
function largest(a, b, c) {  
    return Math.max(a, b, c);  
}
```

```
console.log(largest(15, 45, 30));
```

Output

45

Q3. Create a function to reverse a string.

```
function reverseString(str) {  
    return str.split("").reverse().join("");  
}
```

```
console.log(reverseString("JavaScript"));
```

Output

tpircSavaJ

2. Parameters

Q4. Create a function that calculates Simple Interest.

Formula:

$$SI = \frac{P \times R \times T}{100}$$

```
function simpleInterest(p, r, t) {  
  return (p * r * t) / 100;  
}
```

```
console.log(simpleInterest(10000, 5, 2));
```

Output

1000

Q5. Create a function that greets a user using name and city.

```
function greet(name, city) {  
  console.log(`Welcome ${name} from ${city}`);  
}
```

```
greet("Vicky", "Jaipur");
```

Output

Welcome Vicky from Jaipur

3. Return Function

Q6. Return the square of a number.

```
function square(num) {  
  return num * num;  
}
```

```
let result = square(12);
```

```
console.log(result);
```

Output

144

Q7. Return whether a year is Leap Year or not.

```
function isLeapYear(year) {  
  return year % 4 === 0;  
}
```

```
console.log(isLeapYear(2028));
```

Output

true

Q8. Return the factorial of a number.

```
function factorial(n) {  
  let fact = 1;  
  
  for(let i = 1; i <= n; i++) {  
    fact *= i;  
  }  
}
```

```
    return fact;
  }

  console.log(factorial(5));
```

Output

120

4. Arrow Functions

Q9. Convert Celsius to Fahrenheit using Arrow Function.

```
const fahrenheit = c => (c * 9/5) + 32;

console.log(fahrenheit(25));
```

Output

77

Q10. Find cube of a number.

```
const cube = n => n ** 3;

console.log(cube(4));
```

Output

64

Q11. Find total price after adding GST (18%).

```
const finalPrice = amount => amount + (amount * 18 / 100);
```

```
console.log(finalPrice(1000));
```

Output

1180

Q12. Check whether a string is palindrome.

```
const palindrome = str =>
```

```
  str === str.split("").reverse().join("");
```

```
console.log(palindrome("madam"));
```

Output

true

5. Callback Functions

Q13. Create a Calculator using Callback Functions.

```
function calculate(a, b, operation) {  
  return operation(a, b);  
}
```

```
function add(x, y) {  
  return x + y;  
}
```

```
console.log(calculate(10, 20, add));
```

Output

30

Q14. Use Callback to process student marks.

```
function processMarks(marks, callback) {  
    callback(marks);  
}
```

```
function showResult(marks) {  
    if (marks >= 40)  
        console.log("Pass");  
    else  
        console.log("Fail");  
}
```

```
processMarks(55, showResult);
```

Output

Pass

Q15. Simulate Online Food Delivery.

```
function orderPlaced(callback) {  
    console.log("Order Placed");  
    callback();  
}
```

```
function orderDelivered() {  
    console.log("Order Delivered");  
}
```

```
}
```

```
orderPlaced(orderDelivered);
```

Output

Order Placed

Order Delivered

Mixed Practical Interview Questions

Q16. Count Vowels in a String

```
function countVowels(str) {  
    let count = 0;  
  
    for(let ch of str.toLowerCase()) {  
        if("aeiou".includes(ch))  
            count++;  
    }  
  
    return count;  
}  
  
console.log(countVowels("JavaScript"));
```

Output

3

Q17. Generate Random OTP

```
function generateOTP() {  
  return Math.floor(1000 + Math.random() * 9000);  
}
```

```
console.log(generateOTP());
```

Sample Output

5831

Q18. Find Maximum Number from Array

```
function findMax(arr) {  
  return Math.max(...arr);  
}
```

```
console.log(findMax([10, 45, 22, 80, 65]));
```

Output

80

Q19. Calculate Salary Bonus

```
const calculateBonus = salary => salary * 0.10;
```

```
console.log(calculateBonus(50000));
```

Output

5000

Q20. Higher Order Function Example

```
function operation(a, b, callback) {  
  return callback(a, b);  
}  
  
const multiply = (x, y) => x * y;  
  
console.log(operation(5, 6, multiply));
```

Output

30

Frequently Asked Interview Questions

Q21. Can a function return another function?

```
function outer() {  
  return function() {  
    return "Hello";  
  };  
}
```

```
let result = outer();  
  
console.log(result());
```

Output

Hello

Q22. Can a function be stored in a variable?

```
let greet = function() {  
  console.log("Welcome");  
};
```

```
greet();
```

Output

Welcome

Q23. Can an Arrow Function be used as a Callback?

```
setTimeout(() => {  
  console.log("Executed after 2 seconds");  
}, 2000);
```

Output

Executed after 2 seconds

Lab Practice Question

Create functions for:

1. Check Prime Number
2. Check Armstrong Number
3. Check Sunny Number
4. Check Tech Number
5. Find Factorial
6. Find Fibonacci Series
7. Generate OTP

8. Calculate GST
9. Student Grade System
10. Online Shopping Cart Total

These questions cover **Function Declaration, Parameters, Return Values, Arrow Functions, and Callback Functions** in practical real-world scenarios.

some web based example

Practical Web-Based Examples Using JavaScript Functions

These examples are useful for **college labs, mini projects, interviews, and real websites**.

1. Login Form Validation (Function Declaration)

Question:

Validate username and password before login.

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>Login Form</h2>
```

Username:

```
<input type="text" id="username"><br><br>
```

Password:

```
<input type="password" id="password"><br><br>
```

```
<button onclick="login()">Login</button>
```

```
<h3 id="result"></h3>
```

```
<script>
```

```
function login() {
```

```
    let user = document.getElementById("username").value;
```

```
    let pass = document.getElementById("password").value;
```

```
    if(user === "admin" && pass === "1234") {
```

```
        document.getElementById("result").innerHTML =  
            "Login Successful";
```

```
    }
```

```
    else {
```

```
        document.getElementById("result").innerHTML =  
            "Invalid Credentials";
```

```
    }
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

2. GST Calculator (Parameters + Return)

Question:

Calculate GST and Final Amount.

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>

<input type="number" id="amount" placeholder="Enter
Amount">
<button onclick="calculate()">Calculate</button>

<h2 id="output"></h2>

<script>
function calculateGST(amount) {
    return amount * 0.18;
}

function calculate() {
    let amount =
    Number(document.getElementById("amount").value);

    let gst = calculateGST(amount);

    document.getElementById("output").innerHTML =
        "GST = ₹" + gst +
        "<br>Total = ₹" + (amount + gst);
}
</script>

</body>
</html>
```

3. Dark Mode Toggle (Arrow Function)

Question:

Create a Dark Mode button.

```
<!DOCTYPE html>
<html>
<body>

<button id="themeBtn">Dark Mode</button>

<script>
const changeTheme = () => {
  document.body.classList.toggle("dark");
};

document.getElementById("themeBtn")
  .addEventListener("click", changeTheme);
</script>

<style>
.dark{
  background:black;
  color:white;
}
</style>

</body>
</html>
```

4. Live Character Counter (Arrow Function)

Question:

Count characters while typing.

```
<!DOCTYPE html>
<html>
<body>

<textarea id="msg"></textarea>

<h3 id="count">0 Characters</h3>

<script>
const countChars = () => {
  let text =
    document.getElementById("msg").value;

  document.getElementById("count").innerHTML =
    text.length + " Characters";
};

document.getElementById("msg")
  .addEventListener("input", countChars);
</script>

</body>
</html>
```

5. Digital Clock (Callback Function)

Question:

Display current time every second.

```
<!DOCTYPE html>
<html>
```

```
<body>

<h1 id="clock"></h1>

<script>
function showTime() {
    let now = new Date();

    document.getElementById("clock").innerHTML =
        now.toLocaleTimeString();
}

setInterval(showTime, 1000);
</script>

</body>
</html>
```

Callback Used

```
setInterval(showTime, 1000);
```

showTime is passed as a callback function.

6. Image Slider (Callback + Function)

```
<!DOCTYPE html>
<html>
<body>


```

```
<script>
```

```
let i = 1;
```

```
function changelImage() {
```

```
    i++;
```

```
    document.getElementById("slider").src =
```

```
    "https://picsum.photos/400/200?random=" + i;
```

```
    if(i > 10) {
```

```
        i = 1;
```

```
    }
```

```
}
```

```
setInterval(changelImage, 3000);
```

```
</script>
```

```
</body>
```

```
</html>
```

7. OTP Generator (Return Function)

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<button onclick="showOTP()">Generate OTP</button>
```

```
<h2 id="otp"></h2>
```

```
<script>
function generateOTP() {
    return Math.floor(
        1000 + Math.random() * 9000
    );
}

function showOTP() {
    document.getElementById("otp").innerHTML =
        generateOTP();
}
</script>

</body>
</html>
```

8. Student Grade Calculator

```
<!DOCTYPE html>
<html>
<body>

<input type="number" id="marks"
placeholder="Enter Marks">

<button onclick="checkGrade()">
Check Grade
</button>

<h2 id="result"></h2>
```

```
<script>
function getGrade(marks) {

    if(marks >= 90)
        return "A+";

    if(marks >= 75)
        return "A";

    if(marks >= 60)
        return "B";

    if(marks >= 40)
        return "C";

    return "Fail";
}

function checkGrade() {

    let marks =
    Number(document.getElementById("marks").value);

    document.getElementById("result").innerHTML =
        getGrade(marks);
}
</script>

</body>
</html>
```

9. Search Filter (Arrow Function)

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<input type="text"
```

```
id="search"
```

```
placeholder="Search">
```

```
<ul id="list">
```

```
  <li>Java</li>
```

```
  <li>Python</li>
```

```
  <li>JavaScript</li>
```

```
  <li>PHP</li>
```

```
</ul>
```

```
<script>
```

```
const searchItem = () => {
```

```
  let keyword =
```

```
  document.getElementById("search")
```

```
  .value.toLowerCase();
```

```
  let items =
```

```
  document.querySelectorAll("#list li");
```

```
  items.forEach(item => {
```

```
    item.style.display =
```

```
        item.innerText.toLowerCase()
        .includes(keyword)
        ? "block"
        : "none";

    });
};

document.getElementById("search")
.addEventListner("keyup", searchItem);
</script>

</body>
</html>
```

10. Online Shopping Cart Total

```
<!DOCTYPE html>
<html>
<body>

<input type="number" id="qty"
placeholder="Quantity">

<button onclick="calculateBill()">
Calculate
</button>

<h2 id="total"></h2>

<script>
```

```
function calculateTotal(qty, price) {  
    return qty * price;  
}  
  
function calculateBill() {  
  
    let qty =  
    Number(document.getElementById("qty").value);  
  
    let total =  
    calculateTotal(qty, 500);  
  
    document.getElementById("total").innerHTML =  
        "Total Amount = ₹" + total;  
}  
</script>  
  
</body>  
</html>
```

Most Useful Real-World Function Examples

Project	Function Type Used
Login Form	Function Declaration
OTP Generator	Return Function
GST Calculator	Parameters + Return
Student Grade System	Return Function
Dark Mode	Arrow Function
Search Box	Arrow Function
Digital Clock	Callback Function

Project	Function Type Used
Image Slider	Callback Function
Shopping Cart	Parameters + Return
Form Validation	Function Declaration

These examples closely resemble features used in modern websites such as e-commerce stores, student portals, admin dashboards, and web applications.